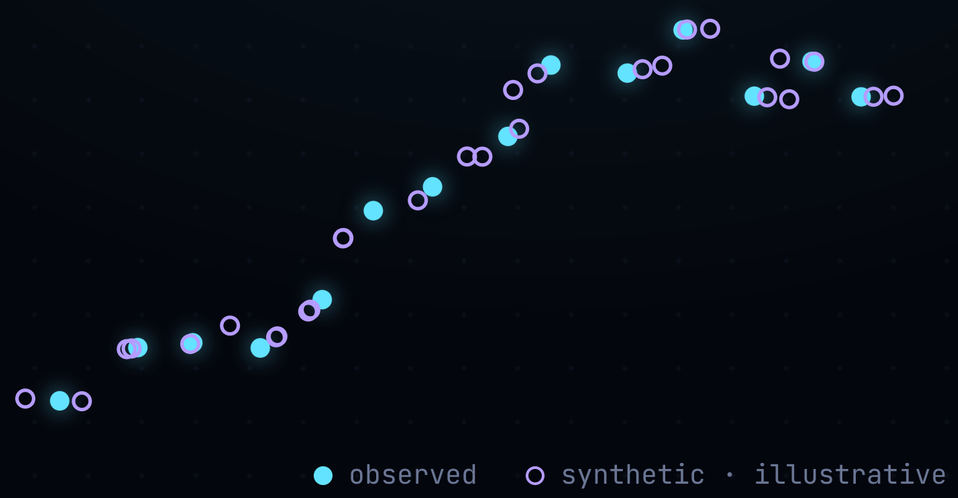
WEEK 5 · FIRST INTERIM PRESENTATION · ECS7029P RESEARCH METHODS · 2026/27

Can we learn more when we cannot collect more data?

Counterfactual Residual Data Augmentation for Regression

Hossein Mohebbi · Oliver Schulte · Ke Li · Pascal Poupart

University of Waterloo · Vector Institute · Simon Fraser University — ICML 2026 · PMLR 306



Proposal stage · no replication results yet

Anik · Alvi · Atul

Small, noisy tables are hard to learn from

01 Few samples

02 Costly to collect more

03 Noisy observations

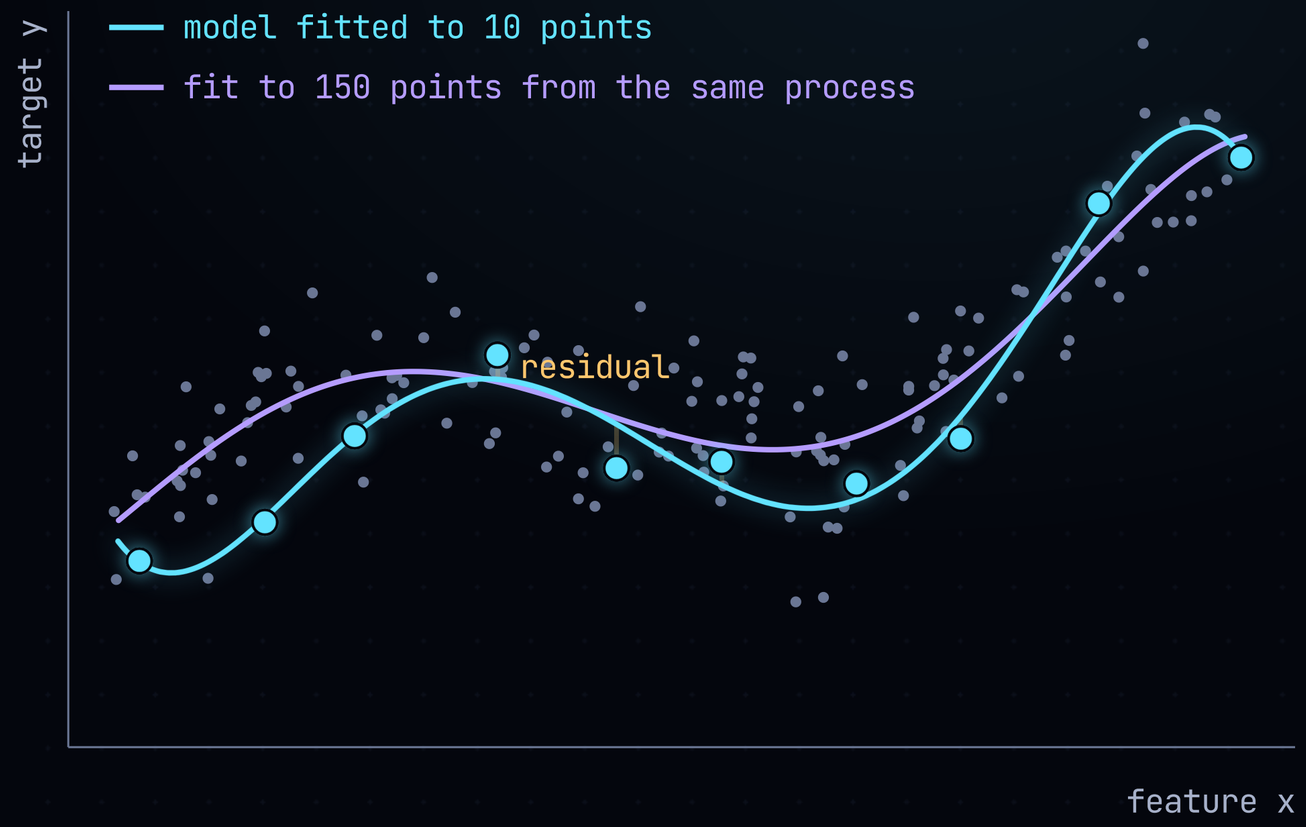
REGRESSION

Predict a *number* from a row of features — e.g. a building's heating load from its geometry.

RESIDUAL

What the model got wrong for that row: observed − predicted.

$$z_i = y_i - \hat{g}(x_i)$$



Illustrative example — synthetic points, real least-squares fits

Images have label-safe transformations. Tables rarely do.

Images: label-preserving for suitable transforms



LABEL “7” ✓ small rotations keep it a 7

...but rotate a 6 by 180° and it reads as a 9

Tabular regression: no known label-safe transform

FLOOR AREA	BEDS	GARAGE	LOCATION	PRICE (TARGET)
92 m²	3	basic	zone B	£285,000
110 m²	3	basic	zone B	£285,000

NAIVE COPY floor area +20%, price copied over

✗ Unjustified label: nothing tells us what this house would sell for

Illustrative example — invented values

INTERPOLATE ROWS

SMOTE-style, C-Mixup (Chawla et al., 2002; Yao et al., 2022)

GENERATE ROWS

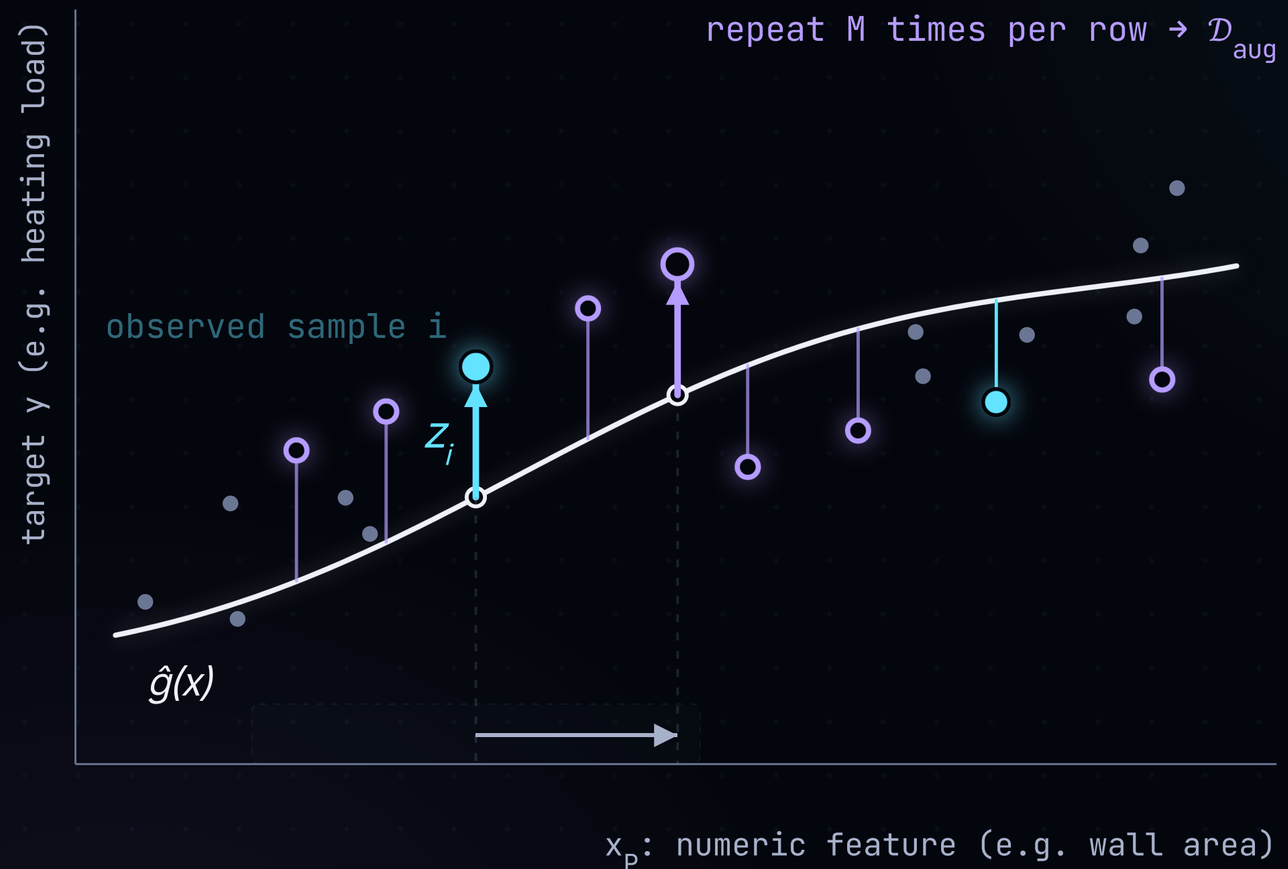
CTGAN, TabDDPM — target treated as just another column (Xu et al., 2019; Kotelnikov et al., 2023)

AUTHORS’ CRITIQUE

can help in places, but don’t keep each row’s *own* residual noise; can raise MSE in their benchmarks

Row: illustrative, invented values · Approaches as characterised by Mohebbi et al. (2026), §2 & Table 2

Move a safe numeric feature. Keep the observed noise.



Illustrative sketch — invented points, not paper data. One feature shown; CRDA may scale several screened numeric features.

Paper's analogy (§1): change a house's garage finish, keep its "bidding-war" residual.

1. **Fit a baseline predictor**
   $\hat{g} \leftarrow \text{MLP or XGBoost on } \mathcal{D}_{\text{train}}$
2. **Compute the original residual**
   $z_i = y_i - \hat{g}(x_i)$
3. **Scale selected numeric features**
   $x'_{i,P} = x_{i,P}\,(1+\delta), \quad \delta \sim \mathcal{U}[-p, p]$
   $x_{i,R}$ held fixed · repo default p = 0.1 · shown: δ = +0.09
4. **Reuse the same residual**
   $y'_i = \hat{g}(x'_i) + z_i$
5. **Repeat M times per row**
   $\mathcal{D}_{\text{aug}} = \{(x'_i, y'_i)\}$

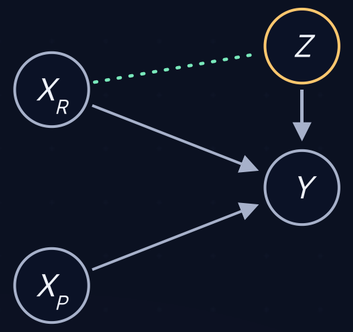
04 — WHAT MUST BE TRUE

It only works if the noise ignores the features we move

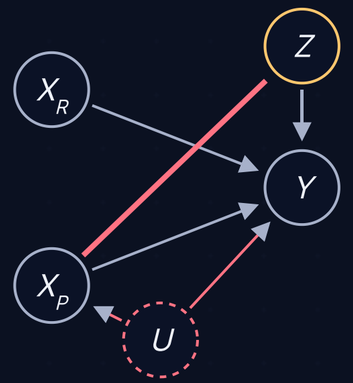
$$P(Z \mid X_P, X_R) = P(Z \mid X_R)$$

Given the fixed features, the residual carries no information about the perturbed ones.

(A) Theory · Assumption 3.1 — assumed, not guaranteed by the algorithm



✓ holds X_P moves Y, not Z. Z may depend on X_R.



✗ breaks hidden U drives X_P and Y; Z absorbs U.

→ causal effect ⋯ allowed dependence ━ violates 3.1 after paper Fig. 1

(B) HEURISTIC SCREENS — EVIDENCE, NOT PROOF

F1 ~~F2~~ F3 F4 ~~F5~~ F6 ~~F7~~ F8 illustrative features

PC CAUSAL-GRAPH CHECK
drop features with an edge to $\hat{z}$

PEARSON SCREEN
drop features correlated with $\hat{z}$ (repo default $|r| \geq 0.1$)

survivors → X_P

(C) PREDICTIVE GATE · WILCOXON SIGNED-RANK
10-fold CV errors, augmented vs. unaugmented
p < 0.05 → use CRDA otherwise → keep baseline
Evidence on validation folds — not a guarantee on new data

Also fails if ĝ is inaccurate: its errors leak into ẑ and stay tied to X_P (§3.3).

Screens are heuristics: PC can miss hidden confounders; Pearson misses non-linear links (§6).

Mohebbi et al. (2026), Assump. 3.1, Fig. 1 adapted, §3.3, §6 · Spirtes et al. (2000) · Wilcoxon (1945)

A focused, checkable reproduction — not a re-invention

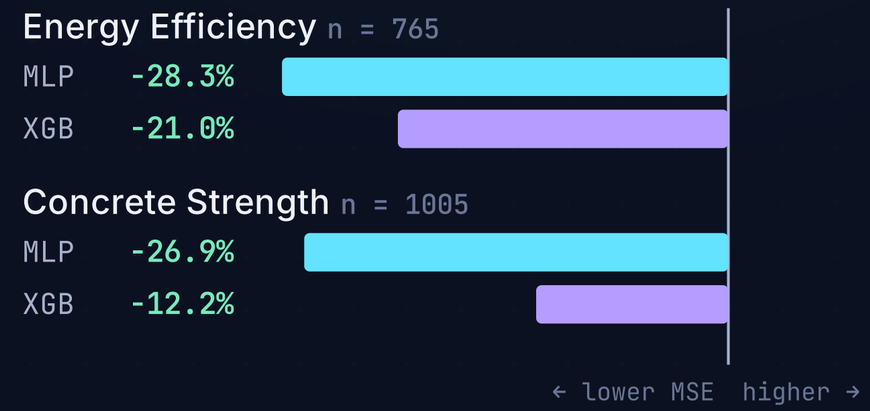
1. **Install & audit**
   authors' repo, Python 3.11, pinned `requirements.txt`
2. **Run the demo**
   `demo/demonstration.ipynb` — Energy Efficiency
3. **Initial scope** PROPOSED
   Energy Efficiency + Concrete Strength · MLP & XGBoost
4. **Controlled comparison**
   unaugmented vs CRDA, identical splits; 3 seeds first pass (paper: 15)
5. **Compare with published**
   our ΔMSE next to Table 1, matching their protocol
6. **Document**
   differences, obstacles, versions, runtimes

PROPOSED PLAN — NOTHING RUN YET

Original authors' reported results — **not our group's reproduced results**

−22.9% MLP **−6.4%** XGBoost

average test-MSE change across 9 benchmark datasets



Paper Table 1 · mean of 15 seeds · largest subset size

CODE-READING QUESTIONS — LINES FOUND IN THE AUTHORS' REPO; WHAT THEY IMPLY IS NOT YET CHECKED

? Table 1 configs set `ignore_filter: true`, so ΔMSE is measured with augmentation forced on — the Wilcoxon decision is only logged
? Code scores the augmented model as the average of 10 CV-fold models; Alg. 1 describes one retrained model

Reported values: Mohebbi et al. (2026), abstract & Table 1 · Code questions: repo experiments/*/config.json, src/experiment.py

Clear ownership, balanced load, every result cross-checked

Proposed roles — confirm with team

A PROPOSED

Anik

Environment & reproducibility

- Python 3.11 env from the authors' pinned requirements
- Run the demo notebook end-to-end
- Seed, config and version logging

CROSS-CHECK

Re-runs one of Alvi's configs independently

SPEAKS

slides 1, 2, 3 + own card

B PROPOSED

Alvi

Experiment execution

- Baseline vs CRDA runs (MLP, XGBoost)
- Controlled splits and seeds
- Saves every config.json / results.csv

CROSS-CHECK

Recomputes Atul's comparison table from raw results

SPEAKS

slides 4, 5 + own card

C PROPOSED

Atul

Analysis, documentation & delivery

- Our ΔMSE next to the paper's Table 1
- Obstacle and deviation log
- Slides, timing and rehearsal

CROSS-CHECK

Rebuilds Anik's environment from the written steps

SPEAKS

slides 6, 8, 9 + own card



Shared by everyone: run at least one config end-to-end · review the final slides · rehearse to time

07 — TARGETS BEFORE WEEK 9

What we will bring to the next update

Proposed targets — weeks and owners to confirm

Week 5 · today — Week 9 · progress update

1 — by Week 6

Environment reproduced

DONE WHEN

Authors' demo notebook runs end-to-end on our machine

EVIDENCE

pip freeze, OS / CPU note, demo output

Anik

2 — by Week 7

Baselines logged

DONE WHEN

Unaugmented MLP + XGBoost on dataset 1, fixed seeds

EVIDENCE

config.json + results.csv per run

Alvi

3 — by Week 8

First CRDA comparison

DONE WHEN

Our ΔMSE next to the paper's Table 1 row

EVIDENCE

Side-by-side table, Wilcoxon p-values

Alvi Atul

4 — Ongoing

Obstacle log

DONE WHEN

Every deviation recorded with its fix or open question

EVIDENCE

Shared log, discussed at Week 9

Atul

OUR RESEARCH QUESTION

Do CRDA's reported gains hold when we re-run the authors' own pipeline — and where do they not?

Weeks and owners proposed, pending team agreement · all milestones planned, none started

REFERENCES

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All URLs checked 19 October 2026